

Development of the back

Vertebrae and their alternating intervertebral discs are one of the main manifestations of body segmentation or metamerism. A chain of segments arranged in sequence allows the overall structure to bend when it is moved by the associated muscles. The original body segments, the somites, are formed by the epithelial paraxial mesoderm (Ch. 12) that is found lateral to the neural tube and notochord in the early embryo. The somites provide the embryonic cell populations for bone and muscle. The vertebrae form between the early body segments by the recombination of portions of the somites on the craniocaudal axis, and the muscles attach to adjacent vertebrae. Each vertebra develops from bilateral origins to form a midline centrum, two lateral arches bearing transverse processes that develop lateral and dorsal to the spinal cord, and a midline fused dorsal portion with a spinous process. Individual vertebrae may be distinguished by modifications of these component parts. The intervertebral discs are of dual origin; the anulus fibrosus develops from the sclerotome and the nucleus pulposus from the notochord.

SEGMENTATION OF PARAXIAL MESODERM

Epiblast cells that ingress through the lateral aspect of the primitive node and the rostral primitive streak (see Fig. 10.3) become committed to a somitic lineage. After passing through the streak, these mesoblastic cells retain contact with both the epiblast and hypoblast basal laminae as they migrate, and for some time after reaching their destination. Their final destination is each side of the notochord, where the cell population, initially presomitic or unsegmented mesenchyme, is referred to as paraxial mesoderm when mesenchyme to epithelial transformation occurs and somites form. Somites will form from cultured presomitic mesoderm with or without the presence of neural tube tissue or primitive node tissue. As well as specifying somitic lineage, the position of ingression of the epiblast informs the specific destination of the cells. Thus, those cells that ingress through the lateral portion of Hensen's node form the medial halves of the somites, whereas those ingressing through the primitive streak approximately 200 µm caudal to the node produce the lateral halves of the somites. The two somite halves do not appear to intermingle.

Bilateral segmentation of the paraxial presomitic mesoblastic populations, which divide into discrete epithelial spheres, occurs as a sequential process along the craniocaudal axis. In avian embryos, a pair of somites is formed every 90 minutes until the full number is obtained. The molecular pathway for this synchronous segmentation has been termed the segmentation clock. It has been identified as a conserved process in vertebrates from fish to mammals, and is based on the rhythmic production of messenger RNAs (mRNAs) for the transcription of genes related to Notch, a large transmembrane receptor, and a number of other factors, including members of the Wnt and fibroblast growth factor (FGF) signalling pathways.

Intrinsically coordinated pulses of mRNA expression appear as a wave within the presomitic mesoblast as each somite forms. As new cells enter the paraxial mesoderm caudally, they begin phases of upregulation of the cycling genes, followed by downregulation of these genes. During each cycle, the most cranial presomitic mesoblast will segment and undergo mesenchyme to epithelial transformation to form the next somite.

Experimental evidence (from chick embryos) shows that newly formed paraxial mesoblast cells undergo 12 such cycles before they finally form a somite (Pourquié and Kusumi 2001). Thus, from ingression through the primitive streak to segmentation into a somite takes approximately 18 hours. As the somite number varies between vertebrate species, it is likely that the rate of somite formation also varies and may be longer in human embryos. Indeed, those vertebrates with elongated bodies and many somites form somites more rapidly relative to their developmental rate than those with shorter bodies, a finding

that supports the concept that somite number is controlled, in part, by species-specific cyclical properties of the presomitic mesoblast (Richardson et al 1998, Gomez et al 2008).

The final determination of somitic boundary formation has not yet been fully elucidated but seems to require a periodic repression of the Notch pathway genes. The caudal presomitic mesoblastic cells are thought to be maintained in an immature state by their production of

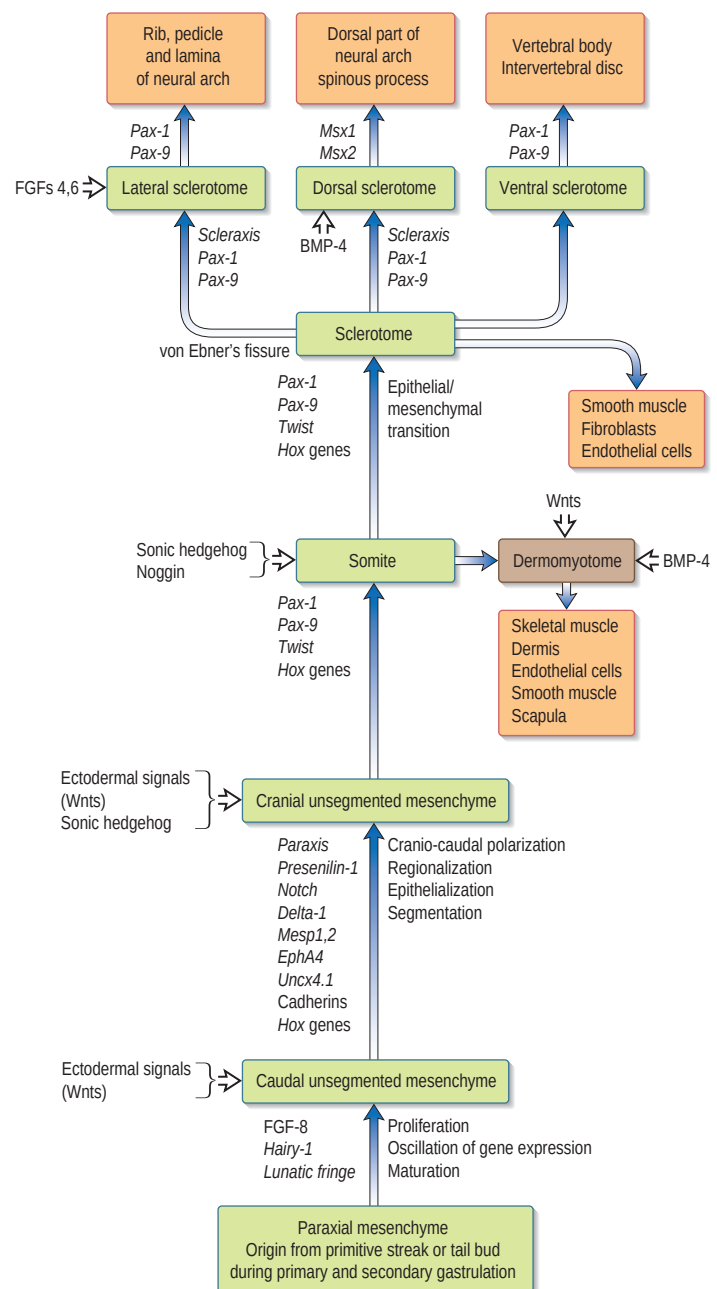


Fig. 44.1 Processes in the development of the paraxial mesenchyme in the avian embryo. Signals are indicated by open arrows; genes are italicized. Abbreviations: BMP, bone morphogenetic protein; FGF, fibroblast growth factor. (Redrawn with permission from Christ B, Huang R, Wilting J 2000 The development of the avian vertebral column. *Anat Embryol* 202: 179–94, Springer.)

FGF8. The cells become competent for segmentation when FGF8 levels drop below a certain threshold. They would then be in close apposition to cells that have segmented (i.e. the next cranial somite).

This area of research is moving rapidly. For an overview of vertebrate segmentation and its clinical implications, see [Pourquié \(2011\)](#). An overview of the processes involved in the development of the paraxial mesoblast, based on the work of [Christ et al \(2000\)](#), is shown in [Figure 44.1](#).

SOMITE DEVELOPMENT

Somitogenesis and somite development encompasses a process in which five main stages can be identified ([Figs 44.2–44.3](#)). A local portion of the most cranial paraxial mesoblast undergoes compaction. The compacted cells undergo a mesenchymal/epithelial transformation, resulting in an epithelial sphere of cells that surrounds free somitocoele cells. This stage marks segmentation of the paraxial mesoderm and somite formation. Shortly after the somite boundaries have been defined, there is a region-specific epithelial/mesenchymal transition of the ventral and ventromedial walls of the somite to form the mesenchymal sclerotome. As the embryo enlarges, the sclerotomal populations on each side become contiguous with the notochord and the neural tube. The rest of the dorsal lateral somitic epithelium remains as the epithelial plate of the somite, also termed the dermomyotome, a proliferative epithelium that will give rise to (nearly) all the striated muscles of the body. Formation of the epaxial myotome begins. Segmentation of the paraxial mesoblast, mesenchymal/epithelial transformation to form epithelial somites and the resultant somite developmental processes (epithelial/mesenchymal transformation to form the sclerotome) all occur in a craniocaudal progression caudal to the otic vesicle from stage 9.

The Golgi apparatus, actin and α -actinin are all located in the apical region of the epithelial somite cells. Cilia develop on the free surface. The cells are joined by tight junctions (a variety of cell adhesion molecule has been demonstrated in epithelial somites). The basal surface, corresponding to the outer circumference of the spherical epithelial somite, rests on a basal lamina that contains collagen, laminin, fibronec-

tin and cytactin. Processes from the somite cells pass through this basal lamina to contact the basal laminae of the neural tube and notochord. Dorsoventral patterning of the putative vertebral column is dependent on the activity of Sonic hedgehog (SHH) from the notochord/floor plate, Wnt1 and Wnt3a from the roof plate, and Wnt6 from the overlying ectoderm.

The sclerotomal population has been subdivided in the chick, and the fates of the cells followed ([Christ et al 2004](#)) ([Figs 44.4–44.6](#)). The



Fig. 44.2 A scanning electron micrograph of a lateral view of an embryo, showing the somites. The cranial somites are at the upper border and the more caudal somites are at the lower border. A change in size of the cranially more advanced somites is apparent. (Photograph by P Collins; printed by S Cox, Electron Microscopy Unit, Southampton General Hospital.)

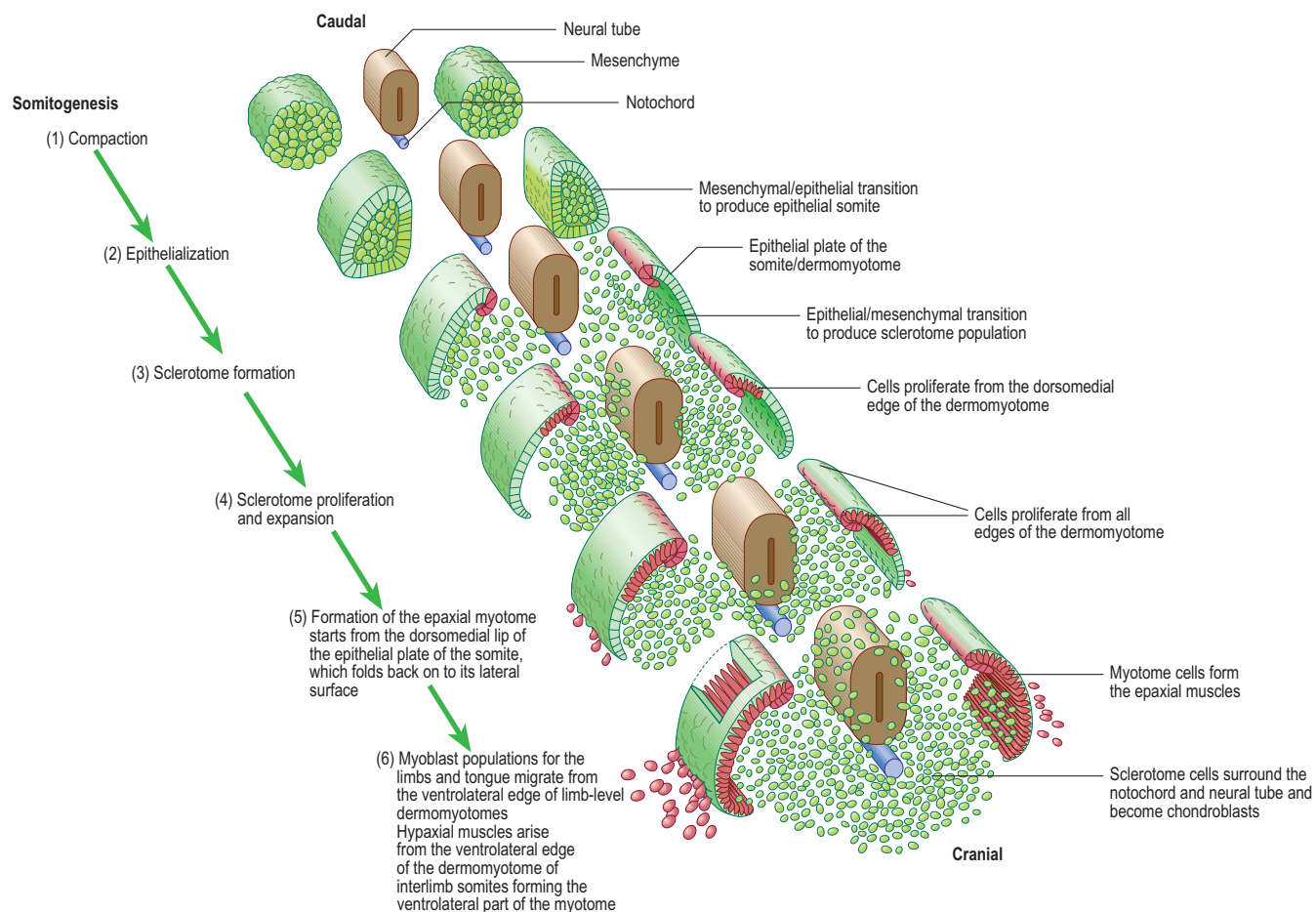


Fig. 44.3 The stages of somite development. Development occurs in a craniocaudal progression. The more cranially placed somites (at the lower right of the figure) are further developed than those caudally placed (at the upper left of the figure). The stages in somite development are given on the left of the figure; more detailed information is given on the right.

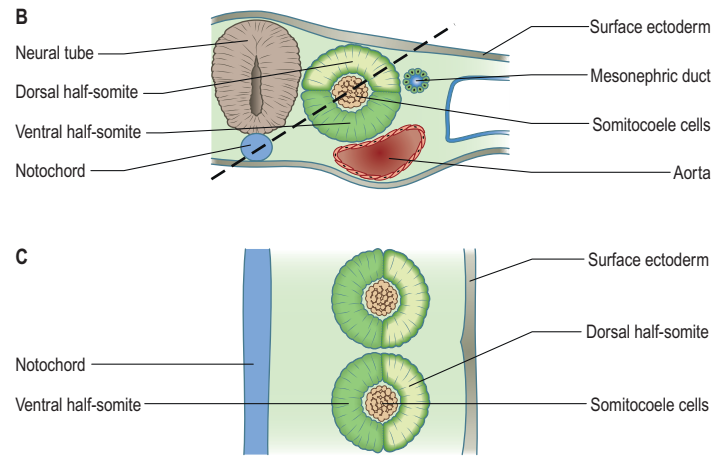
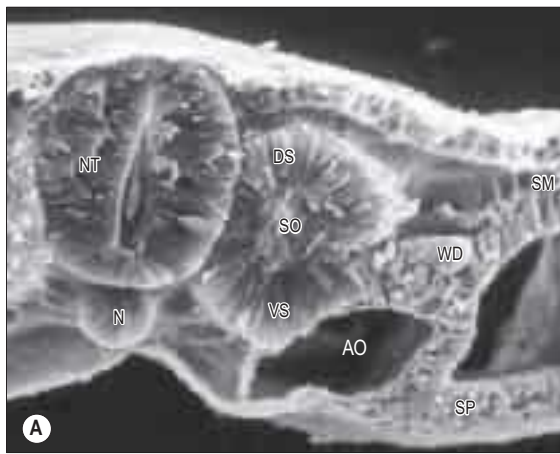


Fig. 44.4 Development of the sclerotome. **A**, A scanning electron micrograph of a transversely fractured 2-day chick embryo. **B–C**, Diagrams of transverse (**B**) and longitudinal (**C**) sections through the somites; the plane in **C** is indicated by the dotted line in **B**. The somites here are in the epithelialization phase of somitogenesis. Abbreviations: AO, aorta; DS, dorsal somite; N, notochord; NT, neural tube; SM, somatopleure; SO, somitocoele; SP, splanchnopleure; VS, ventral somite; WD, Wolffian duct (mesonephric duct). (A, Courtesy of Dr Heinz Jürgen Jacob, Bochum, Germany.)

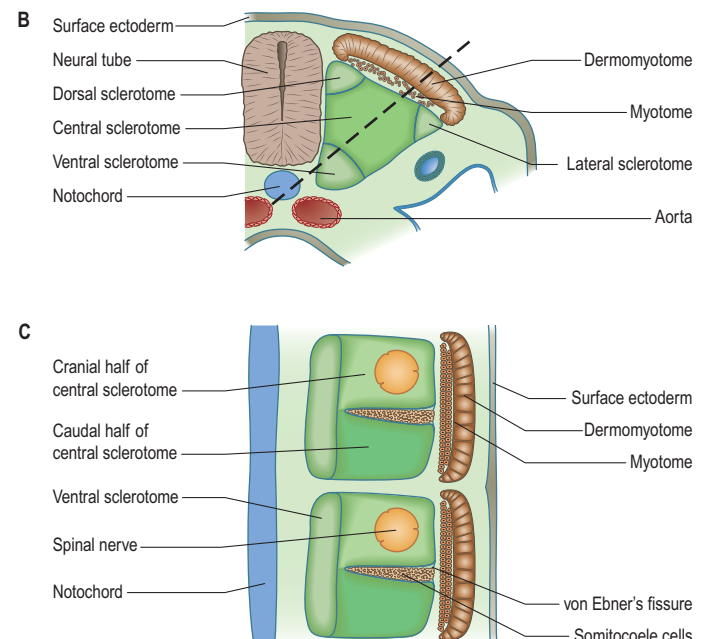


Fig. 44.5 Development of the sclerotome. **A**, A scanning electron micrograph of a transversely fractured 3-day chick embryo. **B–C**, Diagrams of transverse (**B**) and longitudinal (**C**) sections through the developing sclerotome; the plane in **C** is indicated by the dotted line in **B**. Abbreviations: AO, aorta; DM, dermomyotome; N, notochord; NT, neural tube; SC, sclerotome; SM, somatopleure; WD, Wolffian duct (mesonephric duct). (A, Courtesy of Dr Heinz Jürgen Jacob, Bochum, Germany.)

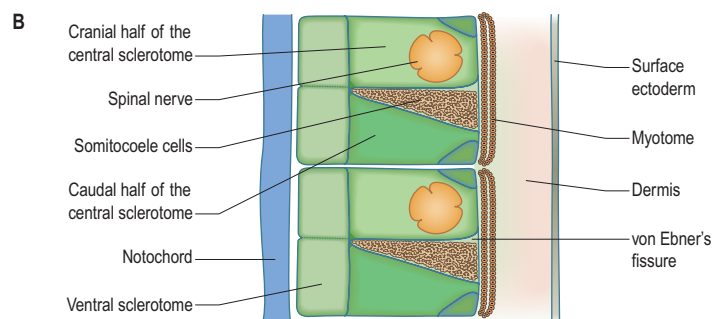
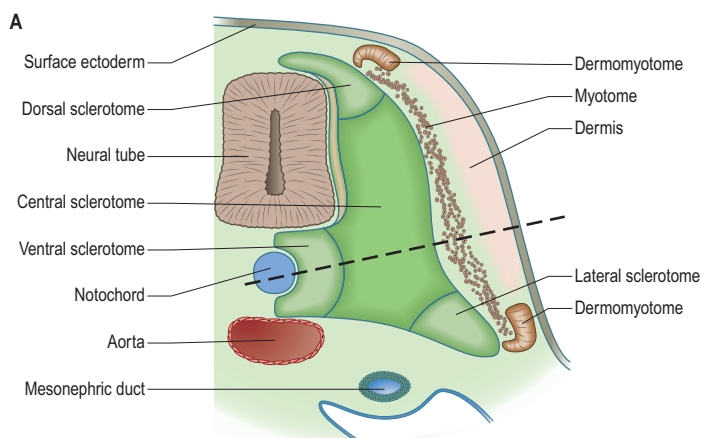


Fig. 44.6 Later development of the sclerotome. **A**, Transverse section. **B**, Longitudinal section through the dotted line indicated in **A**, showing the sclerotomal subdivisions.

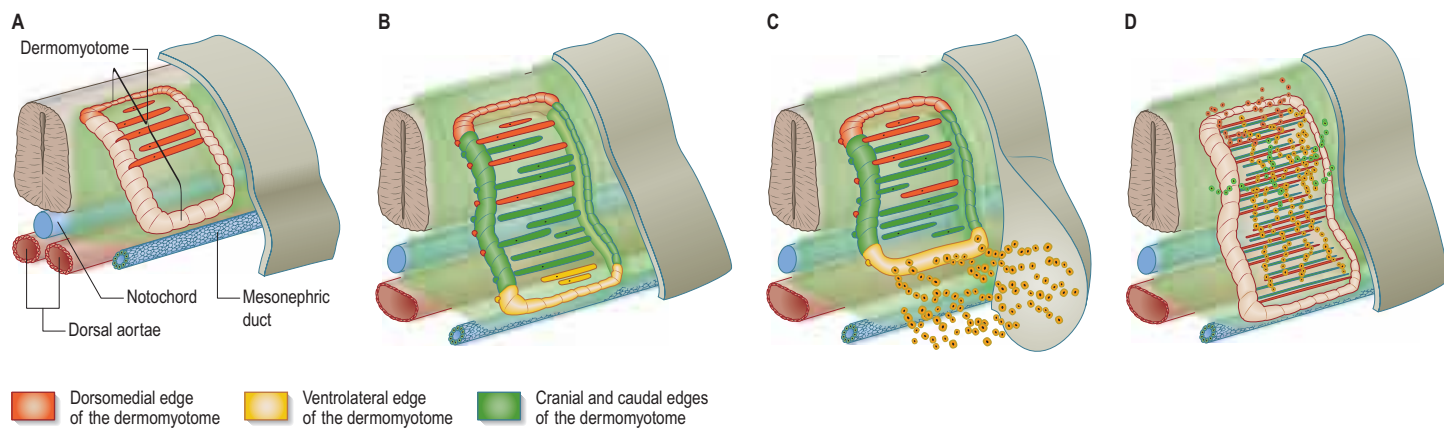


Fig. 44.7 A–D, Epaxial and hypaxial muscle origin from the dermomyotome. **A**, Early dermomyotome proliferation. **B**, The dermomyotome at interlimb level, producing epaxial myoblasts. **C**, The dermomyotome at limb level. Hypaxial dermomyotome cells de-epithelialize from the ventrolateral edge and migrate into the limb bud. **D**, The local dermis is formed later by superficial de-epithelialization of dermomyotome cells.

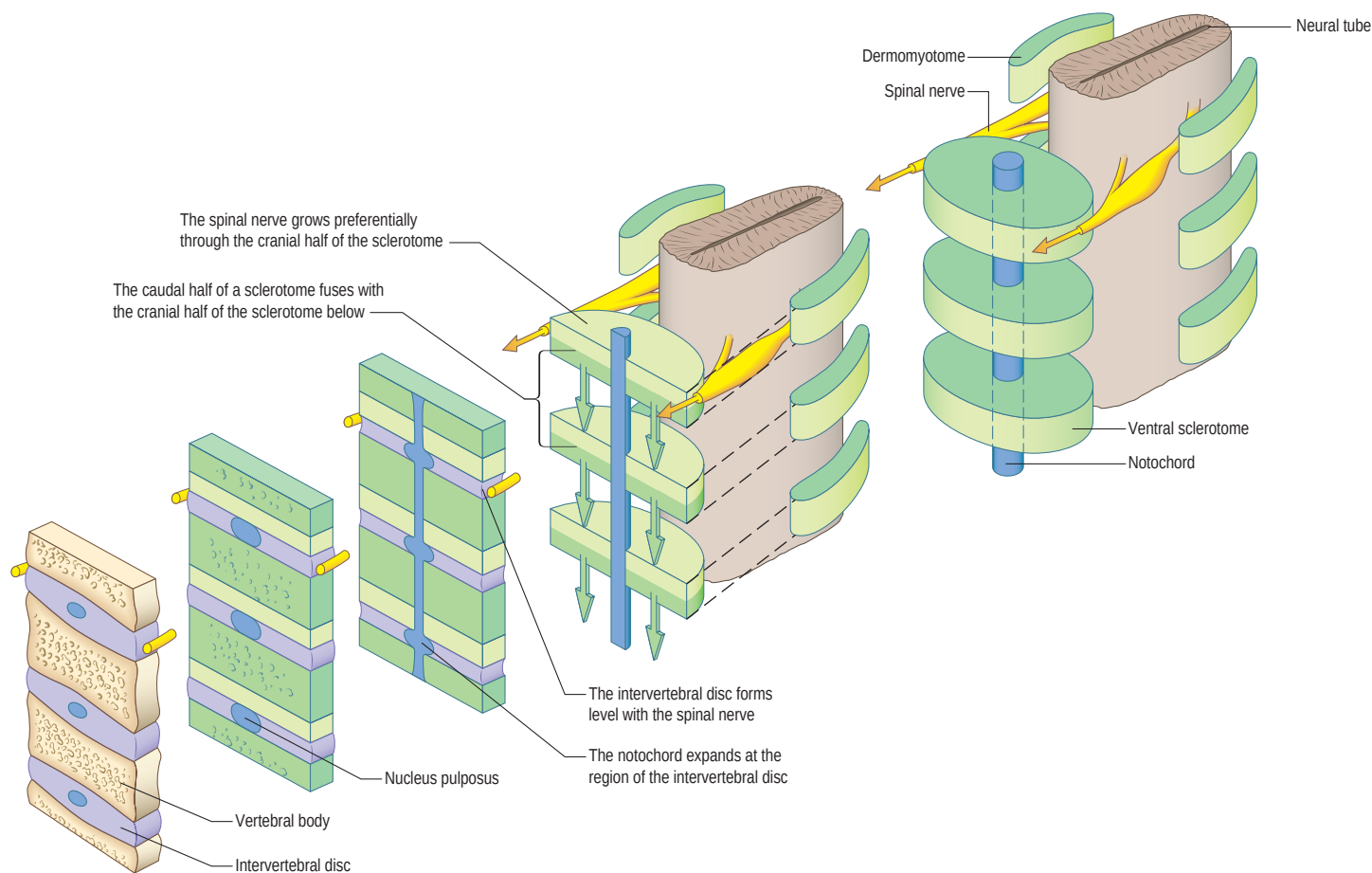


Fig. 44.8 The formation of vertebrae and intervertebral discs from the mesenchymal sclerotomes. Each vertebra is formed from the cranial half of one bilateral pair of sclerotomes and the caudal half of the next pair of sclerotomes.

main mass of the sclerotome is termed the central sclerotome, that portion close to the notochord is termed the ventral sclerotome, and the portions adjacent to the dermomyotome are termed dorsal sclerotome and lateral sclerotome. The central sclerotome remains close to the dermomyotome; it will give rise to the pedicles and ventral parts of the neural arches, and the proximal ribs. The ventral sclerotomal cells, which were always laterally abutting the notochord, proliferate to form an axial cell population within the extracellular matrix of the perinotochordal space, now termed the perinotochordal sheath. The dorsal sclerotomal cells develop relatively late; they invade the space between the surface ectoderm and growing neural tube, and form the dorsal part of the neural arches. The lateral sclerotomal cells give rise to distal ribs and endothelial cells of blood vessels. Generally, there is a dorsolateral expansion of the whole sclerotome rather than the medial migration of a population of sclerotomal cells (Gasser 2006); this can be seen in Figures 44.4–44.6. Sclerotomal cells also give rise to the meninges surrounding the spinal cord, local tendons and ligaments. The somitocoel

cells, which remain mesenchymal throughout somite formation, give rise to the vertebral joints, intervertebral discs and the proximal ribs (Christ et al 2004).

The dermomyotome is a proliferative epithelium that produces cells from four borders (Fig. 44.7). Proliferation at the dorsomedial edge produces cells that elongate from the cranial to the caudal edge of the dermomyotome beneath its apical surface as they move laterally. Cells similarly proliferate from the cranial and caudal edges of the dermomyotome and these cells also elongate across it. The cells that are produced from these three edges are termed the epaxial myotome, and will give rise to skeletal muscle dorsal to the vertebrae, i.e. the epaxial musculature. At limb levels, cells de-epithelialize and migrate from the ventrolateral edges of the dermomyotome into the limb bud. Cells produced from this portion of the occipital somites migrate anteriorly to give rise to the intrinsic muscles of the tongue. At interlimb levels, the hypaxial myotome will give rise to intercostal and abdominal muscles, and extend from the ventrolateral edge of the epithelial

dermomyotome into the body wall as development proceeds (Scaal and Christ 2004).

It was thought that the somite gave rise to segmental portions of the dermis of the skin, as well as bone and muscle. However, it is now clear that the somitic contribution to the skin from the epithelial plate is limited to de-epithelialization of dermomyotomal cells over the epaxial muscles alone, which is a much smaller distribution than the segmental portion of skin usually implied by the term dermatome. The concept that an embryological dermatome, derived from the somite, produces all of the dermis of the skin is therefore outdated.

The regularity of somite formation provides criteria for staging embryos. The staging scheme proposed by Ordahl (1993) will be used in the following account of relative somite development. Ordahl noted that morphogenetic events occur in successive somites at approximately the same rate. The somite most recently formed from the unsegmented mesenchyme is designated as stage I, the next most recent as stage II, and so on. When the embryo has formed an additional somite, the ages of the previously formed somites increase by one Roman numeral. According to this scheme, compaction occurs at stage 0, epithelialization at stage I, formation of mesenchymal sclerotome cells from stage V, and myotome formation from stage VI; early migration of the ventrolateral lip of the dermomyotome and production of myotome cells are still occurring at stage X.

DEVELOPMENT OF SCLEROTOMES

Sclerotomal populations form from the ventral half of the epithelial somite. An intrasegmental boundary (fissure or cleft, sometimes termed von Ebner's fissure), which is initially filled with extracellular matrix and a few cells, appears within the sclerotome and divides it into loosely packed cranial and densely packed caudal halves. The epithelial plate and, later, the dermomyotome span the two half-sclerotomes. The bilateral sclerotomal cell populations migrate towards the notochord and surround it to form the perinotochordal sheath. They undergo a matrix-mediated interaction with the notochord, differentiating chondrogenetically to form the cartilaginous precursor of the vertebral centrum. The perinotochordal sheath starts to express type II collagen, and this is believed to mark a chondrogenic fate, in those mesenchyme cells that contact it. Each vertebra is formed by the combination of much of the caudal half of one bilateral pair of sclerotomes with much of the cranial half of the next caudal pair of sclerotomes. Their fusion around the notochord produces the blastemal centrum of the vertebra (Figs 44.8–44.9). The mesenchyme adjoining the intrasegmental sclerotomic fissure now increases greatly in density to form a well-defined perichordal disc, which intervenes between the centra of two adjacent vertebrae and is the future anulus fibrosus of the intervertebral symphysis ('disc') (see below).

The basic pattern of a typical vertebra is initiated by this recombination of caudal and cranial sclerotome halves (Fig. 44.10), followed by differential growth and sculpturing of the sclerotomal mesenchyme that encases the notochord and neural tube. The centrum encloses the notochord and lies ventral to the neural tube. Condensation of the sclerotomal mesenchyme around the notochord and right and left neural processes can be seen in stage 15 human embryos. The neural processes curve to enclose the neural tube and extend to the dorsolateral angles of the centrum. The neural arch consists of paired bilateral pedicles (ventrolaterally) and laminae (dorsolaterally), which coalesce in the midline dorsal to the neural tube to form the spinous process. On each side, three further processes project cranially, caudally and laterally from the junction of the pedicle and laminae. The cranial and caudal projections are the blastemal articular processes (zygapophyses), which become contiguous with reciprocal processes of adjacent vertebrae; their junctional zones mark the future zygapophysial or facet joints. The lateral projections are the true vertebral transverse processes. Bilateral costal processes (ribs) grow anterolaterally from the ventral part of the pedicles (i.e. near the centrum), from the neighbouring perichordal disc, and, at most thoracic levels, with accessions from the next adjacent caudal pedicles. The costal processes expand to meet the tips of the transverse processes. The definitive vertebral body is compound, and is formed from a median centrum (derived from the cells of the perinotochordal sheath), and bilaterally from the expanded pedicle ends (derived from the migrating sclerotomal populations). These portions of the vertebral body fuse at the neurocentral synchondroses.

The segmental nature of the vertebrae is promoted by the notochord and neural floor plate, which induce the ventral elements of the vertebrae and repress dorsal structures, e.g. the spinous processes (see Fig. 17.7). Excision of the notochord in early embryos results in fusion of the centra and formation of a cartilaginous plate ventral to the neural

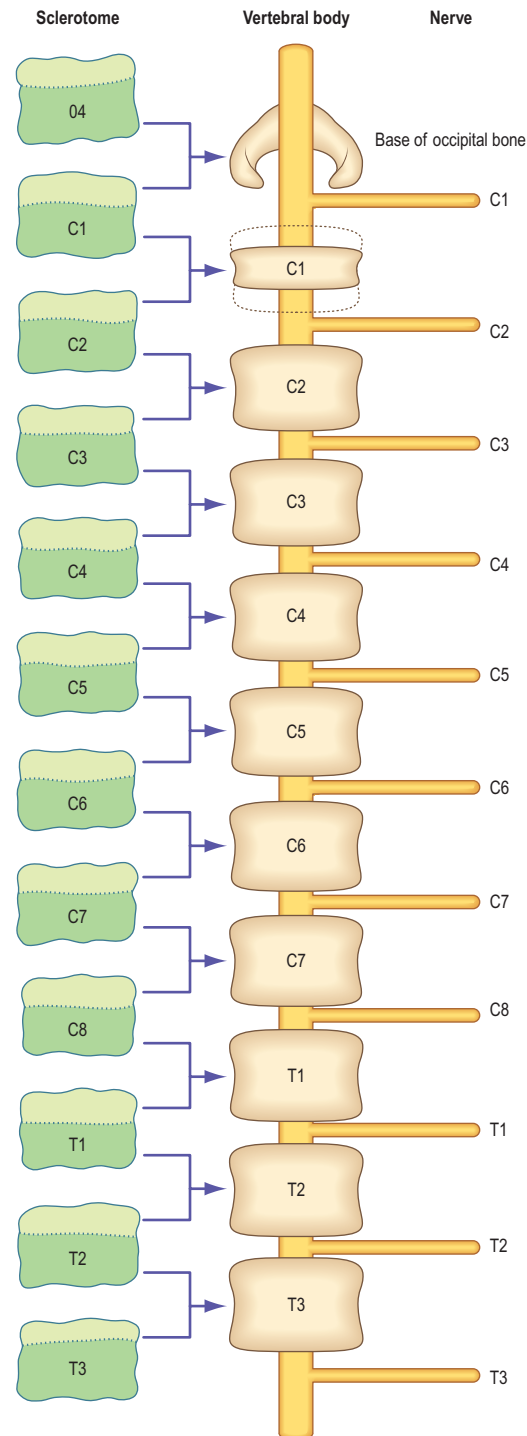


Fig. 44.9 The contribution of the somites to the vertebrae. Each somite induces a ventral root to grow out from the spinal cord. When the sclerotomes recombine, the cranial half of the first cervical sclerotome fuses with the occipital sclerotome above contributing to the occipital bone of the skull. The cervical nerves, beginning with C1, exit above the corresponding vertebra. Nerve C8 exits below the seventh cervical vertebra (C7), and thereafter nerves arise below their numbered vertebrae. (With permission from Larsen WJ 1997 *Human Embryology*, 2nd edn. Edinburgh: Churchill Livingstone.)

tube. Dorsal segmentation is influenced by the spinal ganglia; experimental removal of the ganglia results in fusion of the neural arches and the formation of a uniform cartilaginous plate dorsal to the neural tube.

INTERVERTEBRAL DISCS

Vertebral centra are derived from caudal and cranial sclerotomal halves. An intervertebral disc is formed from the free somitocoel cells within the epithelial somite that migrate with the caudal sclerotomal cells and from notochordal cells. The sclerotomal mesenchyme that forms the

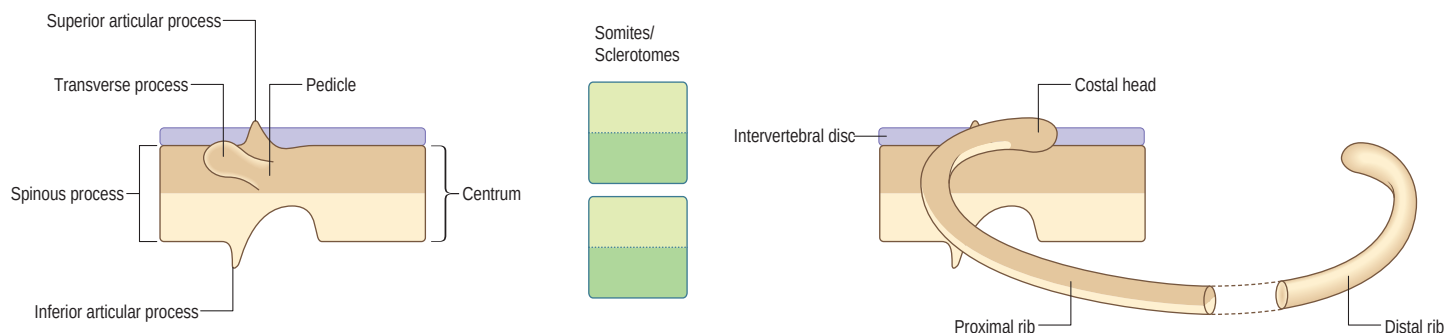


Fig. 44.10 The contribution of two adjacent somites to one vertebra and rib. The intervertebral disc and the costal head are derived from somitocoele cells from one somite, which migrate with the caudal half of the sclerotome. The proximal rib is formed from caudal and cranial somite halves with no mixing of cells; in the distal rib, there is more mixing of cranial and caudal cells as segmentation diminishes in the ventral body wall. (Redrawn with permission from Christ B, Huang R, Wilting J 2000 *The development of the avian vertebral column*. *Anat Embryol* 202: 179–94, Springer.)

centra of the vertebrae replaces the notochordal tissue that it surrounds. In contrast, the notochord expands between the developing vertebrae as localized aggregates of cells and matrix that form the nucleus pulposus of the intervertebral disc (see Fig. 44.8; Fig. 44.11). The intermediate part of each perichordal disc, which forms the anulus fibrosus, surrounds the nucleus pulposus and differentiates into an external laminated fibrous zone and an internal cuff (which lies next to the nucleus pulposus). The inner zone contributes to the growth of the outer. Near the end of the second month of embryonic life, it begins to merge with the notochordal tissue, and is ultimately converted into fibrocartilage. Genetic labelling experiments in the mouse indicate that notochordal cells act as embryonic precursors to all cells found within the nucleus pulposus of the mature intervertebral disc (McCann et al 2012).

The original sclerotomes are coextensive with the individual metameric body segments; each sclerotomic fissure, perichordal disc and maturing intervertebral disc lies opposite the centre of each fundamental body segment. It therefore follows that the discs correspond in level to (i.e. form the anterior boundary of) the intervertebral foramina and their associated mixed spinal nerves, ganglia, vessels and sheaths. Posteriorly, the foramina are bounded by the capsules of the synovial facet joints; the rims of the vertebral notches of adjacent vertebrae lie cranially and caudally. Thus, all the structures listed (and other associated ones) are often designated segmental, whereas vertebral bodies are designated intersegmental because of their mode of development.

(For further discussion of resegmentation theories, the reader is directed to Müller and O’Rahilly 1986, Huang et al 2000, Stern and Vasiliauskas 2000.)

DEVELOPMENT OF VERTEBRAE

The initial movements of sclerotomal cells round the neural tube and the expression of type II collagen signal the blastemal stage of vertebral development (see Fig. 44.11). Chondrification begins at stage 17, initiating the cartilaginous stage. Each centrum chondrifies from one cartilage anlage. Each half of a neural arch is chondrified from a centre, starting in its base and extending dorsally into the laminae and ventrally into the pedicles, to meet, expand and blend with the centrum. By stage 23, there are 33 or 34 cartilaginous vertebrae but the spinous processes have not yet developed, so the overall appearance is of total spina bifida occulta. Fusion of the spines does not occur until the fourth month. The transverse and articular processes are chondrified in continuity with the neural arches. Intervening zones of mesenchyme that do not become cartilage mark the sites of the facet joints and the complex of costovertebral joints, within which synovial cavities later appear.

A typical vertebra is ossified from three primary centres, one in each half-vertebral arch and one in the centrum (see Fig. 43.34). Centres in arches appear at the roots of the transverse processes, and ossification spreads backwards into laminae and spines, forwards into pedicles and posterolateral parts of the body, laterally into transverse processes, and upwards and downwards into articular processes. Classically, centres in vertebral arches are said to appear first in upper cervical vertebrae in the ninth to tenth weeks, and then in successively lower vertebrae, reaching lower lumbar levels in the twelfth week. However, in a radiographic study of unsexed human fetuses (Bagnall et al 1977), a pattern was noted that differed from such a simple craniocaudal sequence. A regular cervical progression was not observed. Centres first appeared in the lower cervical/upper thoracic region, quickly followed by others in the upper cervical region. After a short interval, a third group appeared in

the lower thoracolumbar region and remaining centres then appeared, spreading regularly and rapidly in craniocaudal directions.

The major part of the vertebral body, the centrum, ossifies from a primary centre dorsal to the notochord. Centra are occasionally ossified from bilateral centres that may fail to unite. Suppression of one of these produces a cuneiform vertebra (hemivertebra), which is one cause of lateral spinal curvature (scoliosis). At birth and during the early post-natal years, the centrum is connected to each half-neural arch by a synchondrosis or neurocentral joint. In thoracic vertebrae, costal facets on the bodies are posterior to neurocentral joints.

During the first year, the arches unite first in the lumbar region and then throughout the thoracic and cervical regions. In typical upper cervical vertebrae, centra unite with arches in about the third year, but in lower lumbar vertebrae, union is not complete until the sixth year. The upper and lower surfaces of the bodies and apices of the transverse and spinous processes are cartilaginous until puberty, at which time five secondary centres appear: one in the apex of each transverse and spinous process, and two anular epiphyses (ring apophyses) for the circumferential parts of the upper and lower surfaces of the vertebral body. Costal articular facets are extensions of these anular epiphyses; they fuse with the rest of the bone at about 25 years. There are two secondary centres in bifid cervical spinous processes. Exceptions to this pattern of ossification are described in the appropriate subsections in Chapter 43.

Vertebrae are specified as to region very early in development. If a group of thoracic somites is transplanted to the cervical region, ribs will still develop. It is the sclerotome that is restricted; the myotome will produce muscle characteristic of the new location.

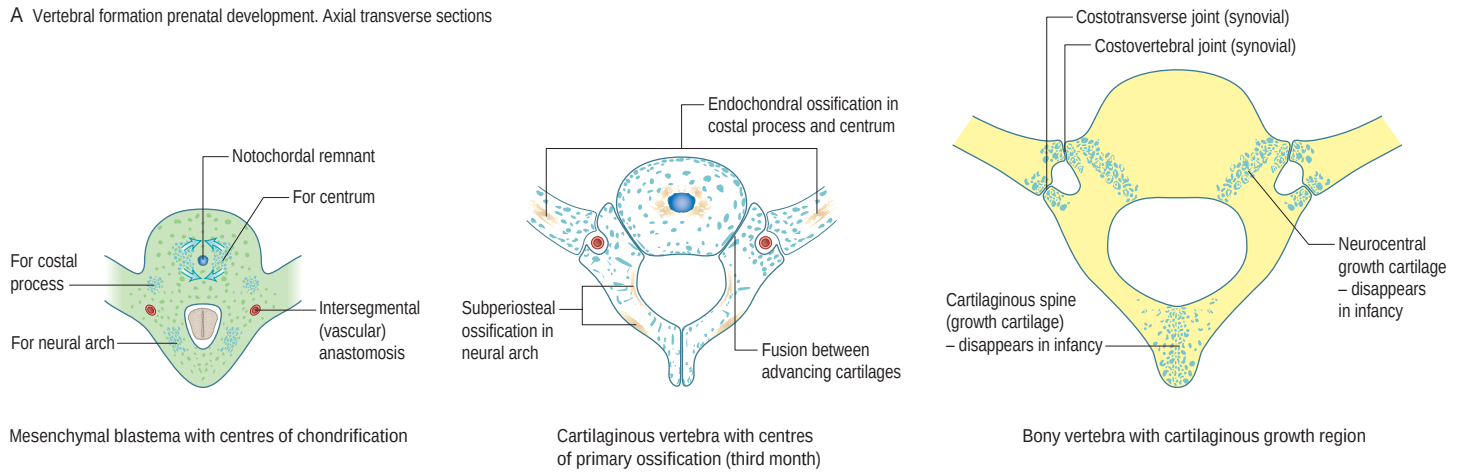
Occipitocervical junction

In humans, the junction between the head and neck (occipitocervical, craniovertebral or spinomedullary junction) is placed at the boundary between the fourth and fifth somites (Müller and O’Rahilly 1994, Muller and O’Rahilly 2003, Hita-Contreras et al 2014). In avian embryos, where all embryonic stages may be obtained experimentally, the occipitocervical boundary has been determined within the fifth somite (Wilting et al 1995). The boundary can first be determined in the human at stage 12 by the observation of hypoglossal nerve rootlets (Fig. 44.12A). At stages 14 and 15, the occipitocervical junction is seen between the hypoglossal rootlets and the first spinal ganglion (Fig. 44.12B–C).

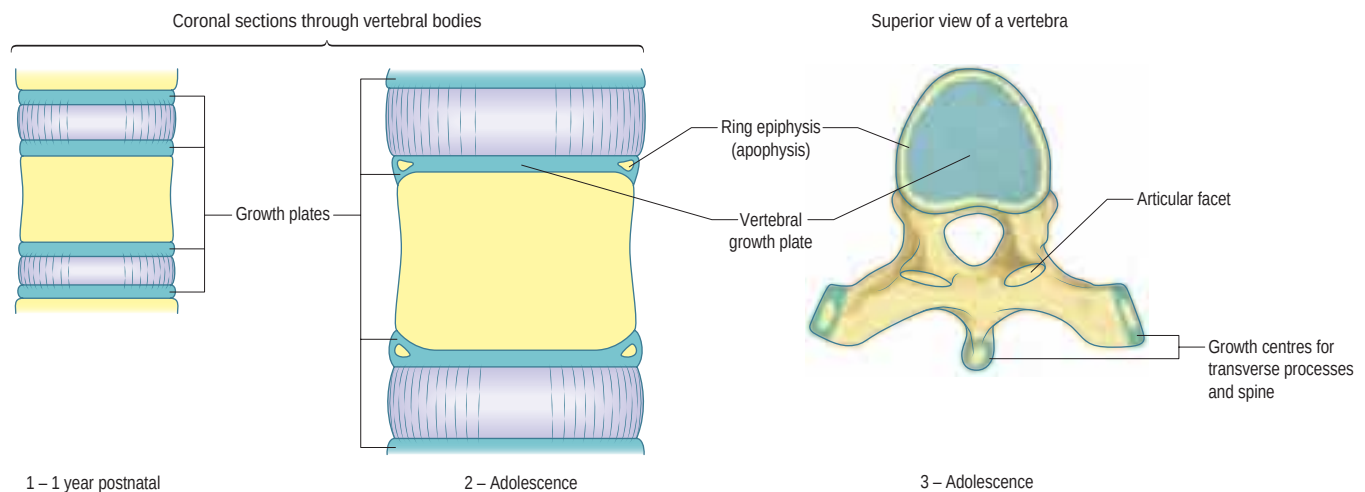
The segmental pattern present in the development of the somites can be seen rostrally in the developing skull base, where mesenchymal condensations equivalent to the centra of occipital somites 2, 3 and 4 are apparent. The first occipital somite disintegrates early and, together with somites 2–4, forms the basiocciput. Occipital sclerotomes 3 and 4 are the most distinct at stage 14, by which time the first three sclerotomes have fused. The hypoglossal rootlets pass through the less dense portion of occipital sclerotome 4, accompanied by the hypoglossal artery. Occipital sclerotome 4 forms an incomplete centrum axially and exoccipital elements laterally. The latter are regarded as corresponding to neural arches and form the rim of the foramen magnum. The occipital condyles develop from the cranial part of sclerotome 5, which is derived from the first cervical somite.

In the occipitocervical junctional region, the centra formed from sclerotomes 5, 6 and 7 have a different fate from those more caudally placed, whereas the lateral portions of these sclerotomes generally develop similarly to those of lower ones. In a study of occipitocervical

A Vertebral formation prenatal development. Axial transverse sections



B Vertebral growth, postnatal development



C Adult vertebrae: derivation

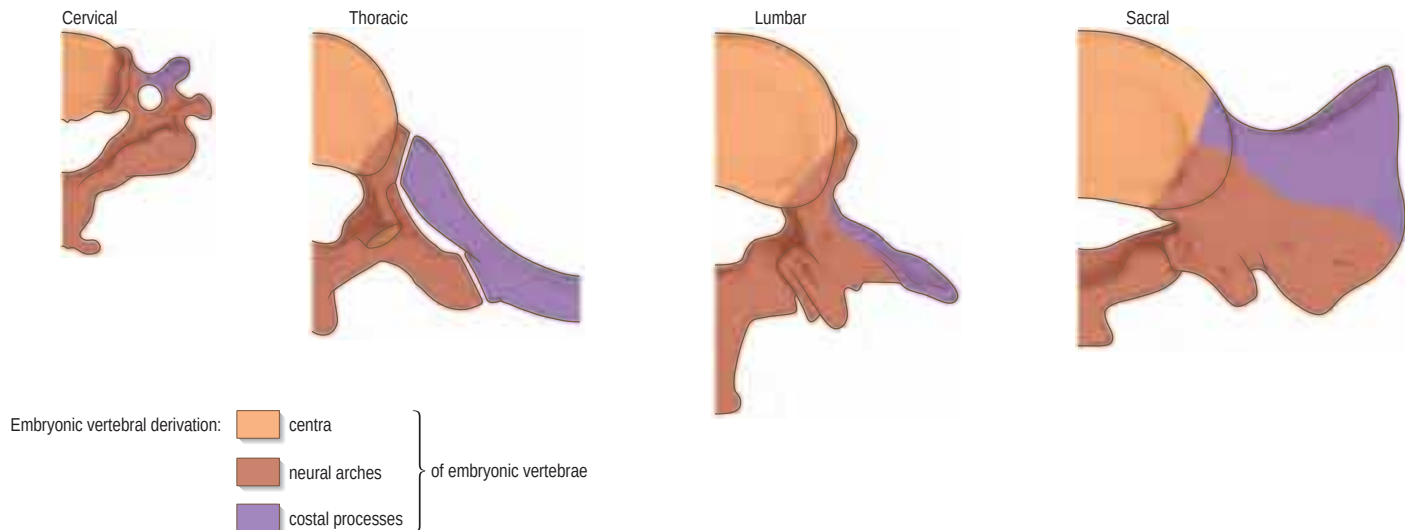


Fig. 44.11 A–B, Vertebral development through blastemal, cartilaginous and pre- and postnatal ossificatory stages. **C**, The derivation of the principal morphological parts of adult vertebrae.

segmentation in human embryos, Müller and O’Rahilly designated the three complete rostral centra that develop in the atlanto-axial region X, Y and Z (Fig. 44.13; Müller and O’Rahilly 1986, Muller and O’Rahilly 2003). They noted that the height of the XYZ complex is equal to that of three centra elsewhere. X is on the level of sclerotome 5, and Y and Z are in line with sclerotome 6 and with the less dense portion of sclerotome 7. During stage 17, a temporary intervertebral disc appears peripherally between Y and Z. It begins to disappear in stage 21, although remains may be found in the adult. No disc develops between X and Y. The origin of the anterior arch of the atlas is unclear. It is evident at stages 21–23 at the level of X or, sometimes, between X and

Y. The posterior arch of the atlas arises from the dense area of sclerotome 5 at the level of X. The XYZ complex belongs to the axis, which means that the atlas does not incorporate a part of the central column (Müller and O’Rahilly 1994). The posterior arch of the axis arises from the dense area of sclerotome 6 and is at the level of Y and Z, particularly Z. XYZ correspond to the three parts of the median column of the axis, where X represents the tip of the dens, Y represents the base of the dens, and Z represents the centrum of the axis. The latter differs from other cervical vertebrae in that it is thicker and square-shaped.

Most defects of the atlas do not contribute to abnormal occipitocervical anomalies and are not associated with basilar invagination.

The bony elements of the occipitocervical junction may be thought of as a central pillar (basiocciput and dens of the axis) surrounded by two rings (the rim of the foramen magnum and occipital condyles superiorly, and the anterior and posterior arches of the atlas inferiorly). Bony anomalies involving this embryologically complex region may produce altered, unstable anatomical relationships that result in the compression of underlying neural and/or vascular structures and may also compromise cerebrospinal fluid dynamics (Menezes 2008, Pang and Thompson 2011, Pang and Thompson 2014). The development of the cervical spine, particularly of the upper cervical vertebrae, is closely related to the development of the basiocciput and exocciput; anomalous development will affect both regions. For example, abnormal prolapse of the dens into the foramen magnum, a condition termed basilar invagination, has been attributed variously to hypoplasia of the basiocciput (clivus), occipital condyles or atlas; deficiencies in the bony arches of the atlas with spreading of the lateral masses; and atlanto-occipital assimilation (Smith et al 2010). Malfusions of the caudal portion of occipital sclerotome 4 and the cranial portion of cervical sclerotome 1 may produce defects of the occipital condyles. The proatlas, a transient bony structure derived from the fourth occipital sclerotome, usually fuses with the three upper occipital sclerotomes to form the occipital bone and the dorsal part of the foramen magnum. Very rarely, remnants of the proatlas persist into adult life; several malformations and anomalies of the most caudal of the occipital sclerotomes have been attributed to proatlas segmentation failures (Menezes and Fenoy 2009, Muhleman et al 2012).

Chiari malformation type I (CM-I) is defined radiographically as a displacement of the cerebellar tonsils of 5 mm or more below the foramen magnum in young adults. Classic CM-I is thought to be a congenital malformation, in which the developing brain is compressed within a hypoplastic posterior fossa, forcing the cerebellar tonsils down through the foramen magnum. Patients with classic CM-I have a characteristically small posterior cranial fossa, particularly below Twining's line (a line joining the anterior tuberculum sellae and internal occipital protuberance), the foramen magnum is constricted transversely, and superior and inferior outlet areas are reduced; the suggested pathogenesis is premature stenosis of the basiexoccipital and exosupraoccipital synchondroses (Milhorat et al 2010). Tonsillar herniation is aetiologically heterogeneous and there are many causes that do not appear to be related to skull-base hypoplasia; CM-I can also occur in association with disorders that appear to be unrelated to skull-base hypoplasia and may be acquired (Tubbs et al 2007).

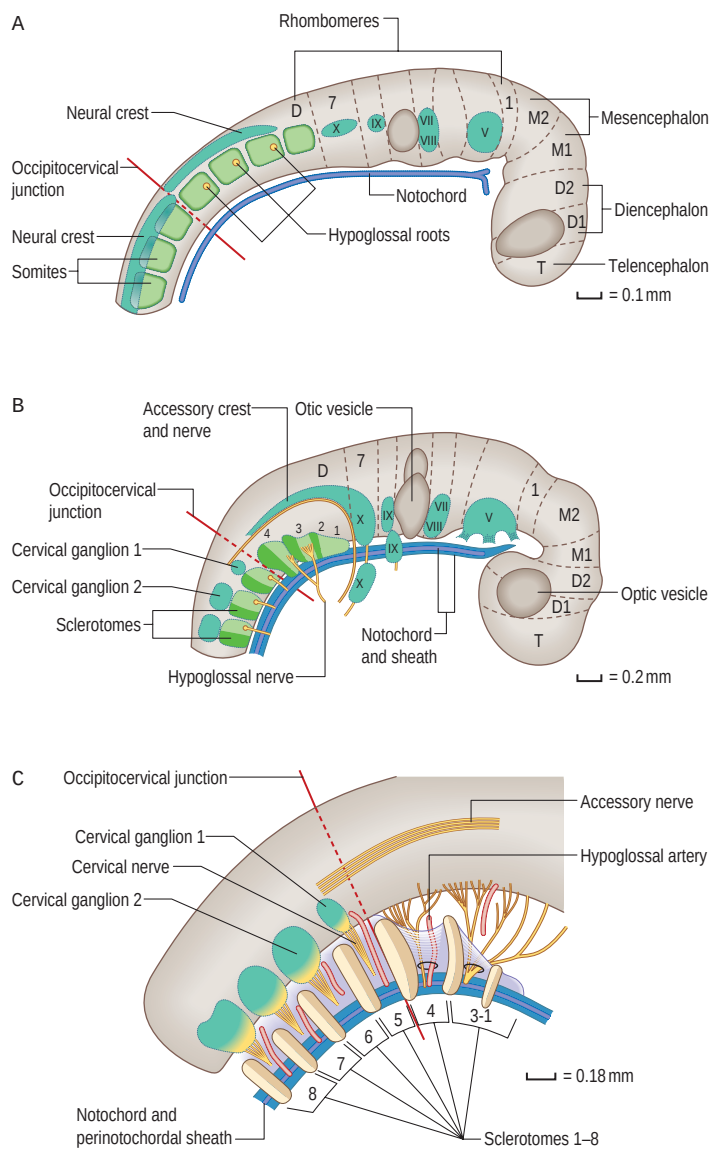


Fig. 44.12 Reconstructions of the occipitocervical region of human embryos. **A**, Stage 12: occipital somites innervated by hypoglossal fibres (small yellow circles). Three cervical somites are shown. The crest-derived ganglia of cranial nerves V, VII, VIII, IX and X are shown in green. Neural crest associated with the occipital somites is hypoglossal and perhaps accessory (also shown in green). **B**, Stage 14: the somites have transformed into sclerotomes and moved ventrally. The less dense cranial and dense caudal parts of the sclerotomes and occipital sclerotomes 1-4 are indicated. Hypoglossal fibres and cervical ventral rami migrate through the less dense parts of the sclerotomes. The occipital neural crest is now seen to be mostly accessory. The cervical crest is subdivided into spinal ganglia. A perinotochordal sheath can be seen extending rostrally to the termination of the notochord. **C**, Stage 15: the dense parts of sclerotomes 1-8 are shown. Intersegmental arteries are visible in the less dense areas of the sclerotomes, as are the spinal nerve fibres. In all diagrams, the occipitocervical junction is indicated by the red line. (After Müller F, O'Rahilly R 1994 Occipitocervical segmentation in staged human embryos. *J Anat* 185: 251-8. Permission from Blackwell Publishing.)

Abnormalities of the axis are usually concerned with fusion of the dens with the centrum of the second cervical sclerotomes. Using the classification of the three complete centra that develop in the atlanto-axial region as X, Y and Z (Müller and O'Rahilly 1986), failure of fusion of X with the YZ complex produces an ossiculum terminale, a dissociated apical odontoid epiphysis. Failure of fusion of the XY complex with Z at the dentocentral synchondrosis, or maintenance of the transitory intervertebral disc at this point, produces an os odontoides, thought to be induced by excessive movement at the time of ossification of the dens (Crockard and Stevens 1995). Hypoplasia and aplasia of the X and Y centra, and aplasia of the Z centrum, will all lead to reduced size of the dens. There are widely differing views about whether this will lead to atlanto-axial instability.

For further details of the development of the human craniovertebral joints and associated ligaments, see Hita-Contreras et al (2014).

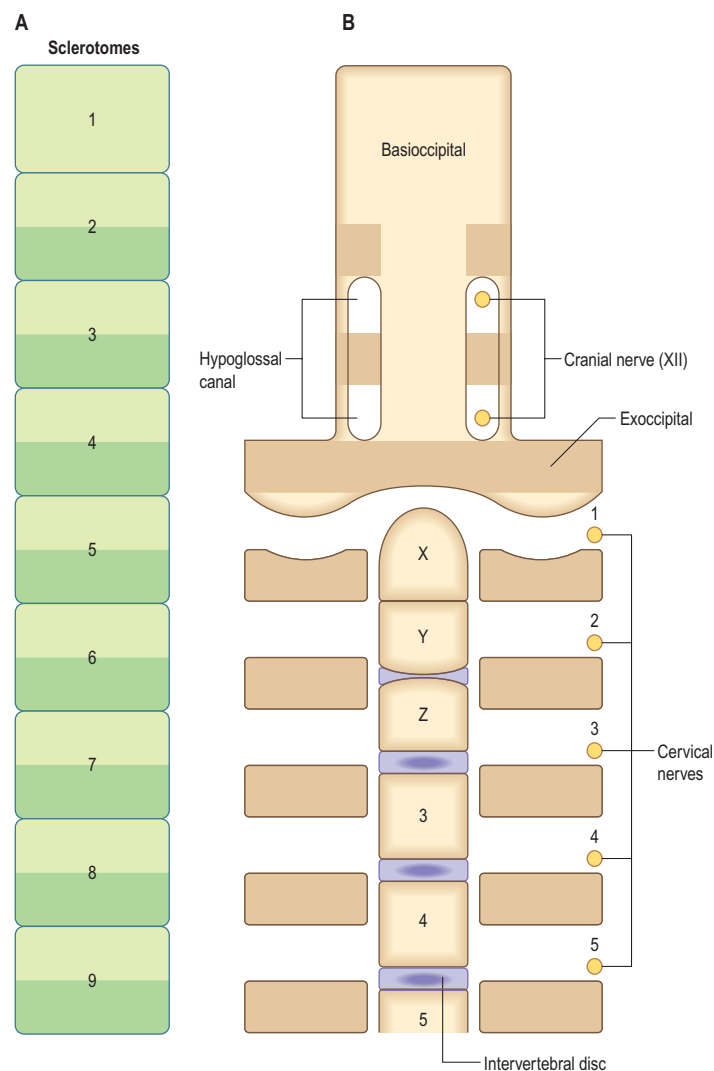


Fig. 44.13 The relationship between the centra and neural arches of the vertebrae and the related spinal ganglia and nerves. **A**, A scheme of the details of the early development of the occipitocervical region. **A**, The column of sclerotomes from occipital somite 1. **B**, A dorsal view of the developing vertebrae, with the centra in the middle and the bilateral components of the neural processes laterally. X, Y and Z are three centra that will produce the tip of the dens of the axis (X), the base of the dens of the axis (Y) and the centrum of the axis (Z). An intervertebral disc appears temporarily between Y and Z during stage 17. No disc develops between X and Y. The occipital condyles are derived from the first cervical sclerotome. (After Müller F, O'Rahilly R 1994 Occipitocervical segmentation in staged human embryos. *J Anat* 185: 251-8. Permission from Blackwell Publishing.)

Third to seventh cervical vertebrae

In cervical vertebrae 3-7 (see Fig. 44.11), the transverse process is dorso-medial to the foramen transversarium. The costal process, corresponding to the head, neck and tubercle of a rib, limits the foramen ventrolaterally. The distal parts of these cervical costal processes do not normally develop; they do so occasionally in the case of the seventh cervical vertebra, and may even develop costovertebral joints. These cervical ribs may reach the sternum.

The seventh cervical vertebra is transitional in shape between cervical and thoracic vertebrae. The laminae are longer than other cervical vertebrae in the neonate and lie almost perpendicular to the basal plane; the inferior articular facets are more upright, and resemble those of thoracic vertebrae; and, in the lateral view, the superior articular facets extend transversely to the top of the transverse processes.

Anomalies of the lower cervical vertebrae are generally caused by inappropriate cervical vertebral fusion; collectively, this is termed Klippel-Feil syndrome. This term includes all congenital fusions of the cervical spine, from two segments to the entire cervical spine. Affected individuals have a low posterior hairline, short neck and limitations of head and neck movement. Scoliosis and/or kyphosis is common.

Thoracic vertebrae

At stage 23, the neural processes of thoracic vertebrae are short, slightly bifurcated and joined by collagenous fibres. The transverse process is prominent. The three facets for articulation with the ribs at the costovertebral and costotransverse joints are present. The thoracic neurocentral and posterior synchondroses are not fused in the neonate; the posterior synchondroses close within 2–3 months of postnatal development and the neurocentral synchondroses are open until 5–6 years of age.

In general, the thoracic spine develops ahead of the cervical and lumbar spine. However, towards the end of the second month, ossification begins in the cartilaginous vertebrae in a craniocaudal progression.

Ribs

Ribs usually develop in association with the thoracic vertebrae. Occasionally, they can arise from the seventh cervical and first lumbar vertebrae.

The costal processes attain their maximum length as the ribs in the thoracic region. Each rib originates from lateral sclerotomal populations, and forms from the caudal half of one sclerotome and the cranial half of the next subjacent sclerotome (see Fig. 44.10). The head of the rib develops from somitocoele cells from one somite, which migrate with the caudal half of the sclerotome. The proximal portion of the rib forms from both caudal and cranial sclerotomal halves; there is no mixing of cells from these origins. The distal portion of the rib forms from caudal and cranial sclerotomal halves; these cells mix as the rib extends into the ventral body wall and segmentation diminishes.

The ribs arise anterolaterally from the ventral part of the pedicles, and form bilateral costal processes that expand to meet the tips of the transverse processes. As they elongate laterally and ventrally, they come to lie between the myotomic muscle plates. In the thorax (see Fig. 44.11), the costal processes grow laterally to form a series of precartilaginous ribs. The transverse processes grow laterally behind the vertebral ends of the costal processes, at first connected by mesenchyme that later becomes differentiated into the ligaments and other tissues of the costotransverse joints. The capitular costovertebral joints are similarly formed from mesenchyme between the proximal end of the costal processes and the perichordal disc, and the adjacent parts of two (sometimes one) vertebral bodies, which are derived from the neural arch. Ribs 1–7 (vertebrosternal) curve round the body wall to reach the developing sternal plates. Ribs 8–10 (vertebrochondral) are progressively more oblique and shorter, only reaching the costal cartilage of the rib above, and contributing to the costal margin. Ribs 11–12 are free (floating), and have cone-shaped terminal cartilages to which muscles become attached.

Lumbar vertebrae

The ribs do not develop distally in lumbar vertebrae (see Fig. 44.11). Their proximal parts become the 'costal processes', while the morphologically true transverse processes may be represented by the accessory processes of the vertebrae. Occasionally, movable ribs may develop in association with the first lumbar vertebra.

Lumbar intervertebral discs are thicker than thoracic discs. By stage 23, the anulus fibrosus can be seen in the peripheral part, and internally, the notochordal cells are expanding to form the nucleus pulposus.

Sacrum

Sacral vertebrae have lower centra and are narrower overall from side to side than their thoracic and lumbar counterparts. Each sacral vertebra is composed of a centrum and bilateral neural processes. The contribution of the costal processes to sacral development was examined by O'Rahilly *et al* (1990). These authors divided the neurocentral junctional area into two parts: anterolateral or alar, and posterolateral. They found the alar element in sacral vertebra 1 to be novel, since it was absent in lumbar vertebra 5. There is support for this view if the course of the dorsal rami of the spinal nerves is used to distinguish the costal elements ventrally from the transverse elements dorsally. The alar elements of the sacral vertebrae are ventral to the sacral dorsal rami, and both costal and transverse portions are posterolateral. The alar element of S1 and S2 forms the auricular surface of the sacrum. At stage 23, the cartilaginous sacral vertebrae have joined and the outline of the future bone can be recognized. Individual pedicles and laminae are very small and can be detected in S3–5.

Ossification of the vertebral column proceeds in a craniocaudal direction. After 16 weeks, it has progressed to L5. Ossification of each

additional vertebra occurs over a period of 2–3 weeks; S2 is ossified by 22 weeks.

Very rarely, significant malformation of the sacral or lumbosacral vertebrae may develop, often in association with a maternal history of diabetes. When there is sacral agenesis, motor paralysis is profound below the affected vertebral level, whereas the sensory disturbance does not relate to the vertebral level so clearly and sensation may be present down to the knees. Bladder involvement is a consistent feature.

SPINA BIFIDA

Spinal dysraphism is a generic term covering a wide range of developmental abnormalities of the vertebral column and spinal cord, characterized by a failure of fusion of midline structures (Sepulveda *et al* 2012). The spectrum of defects includes non-fusion of the entire neural tube with no vertebral arch development (craniorachischisis); non-fusion of the rostral portion of the neural tube with no calvarial or occipital development (anencephaly); and non-fusion of caudal portions of the neural tube and local failure of vertebral arch development (spina bifida) (see Fig. 17.8; Figs 44.14–44.16). Spina bifida may be obvious at birth or before (e.g. spina bifida cystica and spina bifida aperta), or not obvious at birth (spina bifida occulta). Spina bifida cystica occurs where the meninges have developed adjacent to or over the defective neural tissue. Local accumulation of cerebrospinal fluid may push a defective neural plaque or spinal cord superficial to the level of the vertebrae, so forming a meningocele. Alternatively, if one or two spinous processes are absent, a meningeal sac may protrude in the midline above a formed spinal cord, producing a meningocele. In spina bifida aperta, the exposed spinal cord has no cystic covering, although it may be covered by thin membranes. Meningocele occurs in thoracolumbar, lumbar or lumbosacral regions; sacral lesions are less common. The vertebral lesion usually extends cranially further than the neural lesion, showing deformities of the vertebral bodies and laminae. Vertebrae may be wedge-shaped or hemivertebrae, and ribs may be fused or absent. In spina bifida occulta, dysraphism affects primarily the vertebrae; the neural and meningeal elements are not herniated to the surface and so the underlying defects are masked by the skin. This closed form of spina bifida affects about 5% of the population. It usually occurs in the lumbosacral or sacral segments; affected individuals have bifid spinous processes often at the lumbosacral junction and some have a pigmented naevus, angioma, hirsute patch, dimple or dermal sinus on the overlying skin.

Prior to antenatal diagnosis of spina bifida by ultrasonography, most live births with spina bifida cystica had a meningocele. Prenatal diagnosis of meningocele and termination of affected fetuses have led to a significant decrease in the incidence of live births with this condition.

Tethering of the cord and diastematomyelia

The greatest change of vertebral length occurs within the last 6 months of intrauterine life (Ch. 17). When there is any degree of spina bifida, the upward migration of the spinal cord, which normally occurs within the vertebral canal during growth, is limited, and the cord is said to be tethered. Neurological deficits caused by this condition in neonates

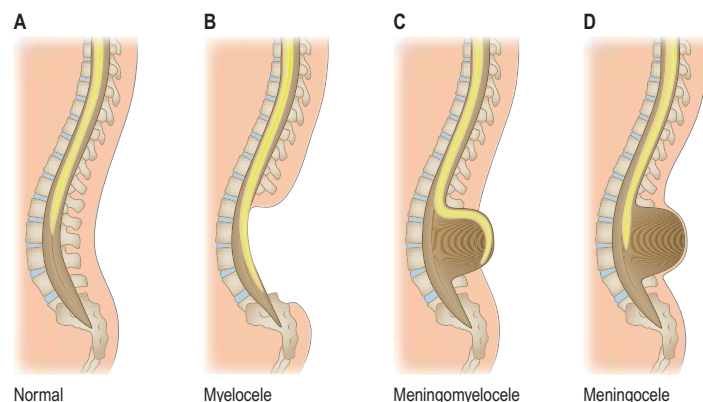


Fig. 44.14 The spectrum of spina bifida. A, Intact neural tube. B, Open spina bifida – myelocele. C, Open spina bifida – myelomeningocele. D, Closed spina bifida with meningocele. (Redrawn with permission from Tulio G, *et al*, Neural Tube Defects: Craniorachischisis and Spina Bifida, in *Obstetric Imaging* (Copel J (ed), Expert Radiology, Series, 2012).)

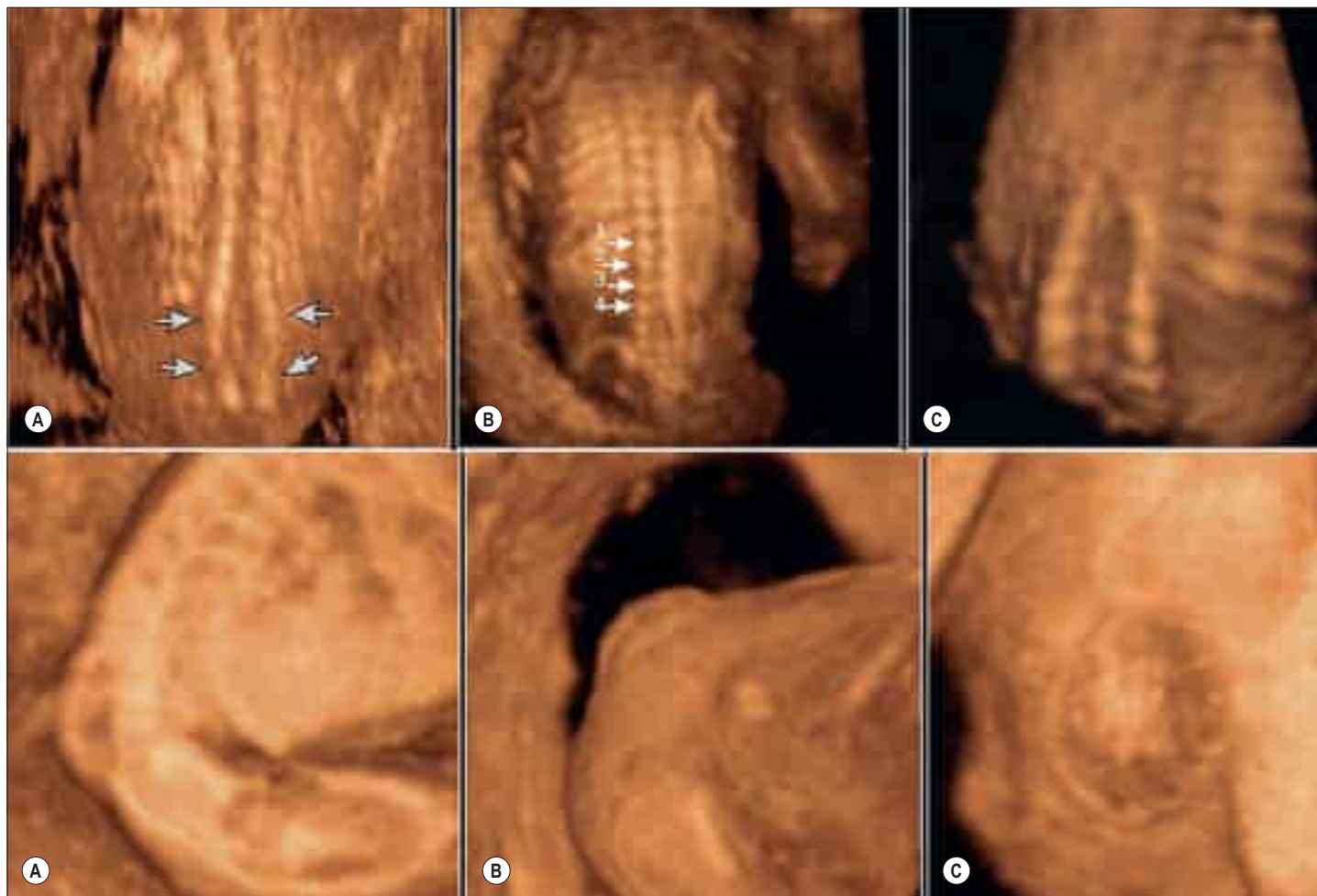


Fig. 44.15 Fetal myelomeningocele. Upper panel: three-dimensional ultrasound maximum transparency mode shows the spinal defect in the coronal (A) and oblique (B) views. Arrows in (A) depict the vertebral dysraphism. The level of the defect can be clearly established (numbers 1 to 4 in (B) show the lumbar vertebrae). Lower panel: three-dimensional ultrasound surface rendering mode shows the spinal defect in the sagittal (A and B) and coronal (C) views. (With permission from Sepulveda W, Ximenes R, Wong AE et al 2012 Fetal magnetic resonance imaging and three-dimensional ultrasound in clinical practice: applications in prenatal diagnosis. Best Pract Res Clin Obstet Gynaecol 26:593–624.)



Fig. 44.16 Fetal magnetic resonance image shows a lumbar myelomeningocele in the sagittal section. (With permission from Sepulveda W, Ximenes R, Wong AE et al 2012 Fetal magnetic resonance imaging and three-dimensional ultrasound in clinical practice: applications in prenatal diagnosis. Best Pract Res Clin Obstet Gynaecol 26:593–624.)

DEVELOPMENT OF THE BACK

become apparent soon after birth, although some individuals show symptoms later in life, especially degrees of urinary dysfunction. Very rarely, tethering symptoms may be associated with abnormal development of the vertebral centra, e.g. when a midline cartilaginous or osseous spicule or a fibrous septum projects into the vertebral canal. These obstacles may split the spinal cord or intraspinal nerve roots into two columns, a condition termed diastematomyelia. Usually, patients with this condition have a cutaneous abnormality, such as a dimple, pigmented naevus or patch of hair, along their back at the level of the tethering.

Ultrasound antenatal imaging of the back

Antenatal ultrasound assessment of the fetal spine at 18–20 weeks routinely assesses the three ossification centres of each vertebra (in the centrum and bilaterally at the lamina–pedicle junction), in the axial, coronal and sagittal planes. In the coronal plane, the posterior ossification centres have a characteristic ‘railtrack’ appearance; the twelfth rib is adjacent to the twelfth thoracic vertebra; the first sacral vertebra is level with the top of the iliac wing. Longitudinal scans of the fetal spine can demonstrate scoliosis and hemivertebrae.

In spina bifida, the posterior ossification centres show abnormal widening in the axial plane, forming a U or V shape; they point outwards from each other and are no longer parallel. Anomalies of the spine (spina bifida) are among the most common fetal malformations noted on antenatal ultrasound examination. The Northern Congenital Abnormality Survey reported a total prevalence at birth and termination of 17.9/10,000 between 1984 and 1996 in the UK and Ireland, in comparison to the prevalence in Europe of 11.5 cases/10,000 over the same time period. For further details, consult [Weston \(2014\)](#).

DEVELOPMENT OF DERMOMYOTOMES

The dermomyotome, which forms from the dorsal half of the early somite, gives rise to all the skeletal muscle in the trunk and limbs and to the intrinsic muscles of the tongue. Muscles in the head arise from the unsegmented paraxial mesoderm rostral to the occipital somites. The dorsomedial lip (border) of the dermomyotome is an infolding of the epithelial plate of the somite and becomes a proliferative epithelial site from which mediolateral growth of the dermomyotome itself occurs. Three structures are required for myogenesis: the neural tube, the notochord/neural floor plate complex and the (early) dorsally placed ectoderm. There is a balance between levels of Shh from the notochord and Wnts from the dorsal neural tube and ectoderm, which promotes myogenesis; high levels of either alone lead to sclerotomal or non-myogenic dermomyotomal development ([Scaal and Christ 2004](#)). Myogenic determination factors, MyoD, myogenin, Myf 5 and herculin/MRF 4, can first be detected in the medial half of the somite as early as stage II, several hours prior to the onset of myotome formation.

The mode of formation of mononucleated, postmitotic, primitive myotubes from the dermomyotome has been elucidated in the chick but not yet confirmed in mammals ([Gros et al 2004](#)) ([Fig. 44.17](#)). In this model, all four borders of the dermomyotome give rise to myotomal cells, produced in two phases. First, cells are produced from the dorsomedial dermomyotomal lip by direct ingression and bidirectional extension. Second, myotomal precursor cells are released by the caudal dermomyotomal border, followed, in turn, by the cranial dermomyotomal border, and lastly, by the ventrolateral dermomyotomal lip. Those myotomal cells arising from the dorsomedial dermomyotomal border contribute cells exclusively to the epaxial domain, while those from the ventrolateral border contribute cells exclusively to the hypaxial domain. The cells arising from the cranial and caudal borders move into both epaxial and hypaxial domains ([Scaal and Christ 2004](#)) (see [Fig. 44.7](#)). Later in development, proliferative muscle progenitor cells originating from the de-epithelializing dermomyotome contribute to fetal muscle growth ([Gros et al 2005](#)).

EPAXIAL MUSCLE: DORSAL TRUNK MUSCLES

Epaxial muscles arise from the dorsomedial lip of the dermomyotome. Early myotome cells are postmitotic embryonic myoblasts; later in development, they fuse to form syncytia, which produce the intrinsic muscles of the back. They bridge the early segmental borders, joining vertebrae together. The epaxial muscles are innervated by the dorsal

rami of each pair of spinal nerves. At later stages, satellite cells originating from the dermomyotome enter the myotome. The mechanism of development of endo-, peri- and epimysium in relation to the epaxial muscles has not been conclusively determined.

HYPAXIAL MUSCLE: VENTROLATERAL TRUNK MUSCLES

Two modes of production of hypaxial musculature are seen, according to axial level (see [Fig. 44.7](#)). At cervical and trunk levels, myotubes are produced by the ventrolateral edge of the dermomyotome in a manner similar to the production of epaxial muscles. As development proceeds, lateral growth of the dermomyotome means that the ventrolateral border extends ventrolaterally into the developing lateral and ventral body wall. This process has been shown to depend on the homeobox-containing transcription factor Pitx2 ([Eng et al 2012](#)). The somite structures that invade the somatopleure of the abdominal and thoracic walls thus consist of two layers, dermomyotome and myotome, that give rise to the muscle blastemata for the intercostal and oblique muscles, transversus abdominis and rectus abdominis. At this time, the number of somatopleural fibroblasts situated within the muscle-forming zone increases, and myotubes can be first seen. There is a subsequent ventral shift of the already separated muscle blastemata within the growing abdominal wall as they attain their definitive positions. Muscle differentiation continues and muscular connective tissue, tendons and aponeuroses develop.

At occipital and limb levels, a ventrolateral dermomyotomal lip is not formed in the same way; instead, cells undergo epithelial/mesenchymal transformation and migrate to their destination. Myoblasts move from the occipital dermomyotomes into the developing mandibular processes to form the intrinsic muscles of the tongue, whereas those opposite the early limb buds migrate into the limbs as dorsal and ventral muscle masses that give rise to the shoulder, hip and appendicular muscles. (For a review of dermomyotomal development, see [Scaal and Christ \(2004\)](#).)

OTHER STRUCTURES DERIVED FROM THE SOMITES

Dermal precursors arise from the central portion of the dermomyotome and, possibly, also from the dorsomedial dermomyotome lip (see [Fig. 44.7](#)). The cells undergo an epithelial/mesenchyme transformation from the dermomyotome and migrate to the dorsomedial subectodermal space overlying the dorsal neural tube. Their transformation seems to be controlled by factors from the neural tube. There is a sharp boundary between dermis that is somite-derived and that which is derived from the somatopleuric mesenchyme of the lateral plate (and which covers the limbs, part of the lateral and all of the ventral body wall).

Somitic cells are now considered to give rise to smooth muscle cells within and around the developing somites, and within the tunica media of the descending aorta ([Wiegrefe et al 2007](#)). All compartments of the epithelial somite, including the somitocoele cells, give rise to angioblasts. The early ventral half of the somite gives rise to the endothelium of ventrolateral blood vessels. Angioblastic cells arising from the dorsomedial dermomyotome migrate mainly into the dorsal dermis, whereas those arising from the dorsolateral part move to the ventrolateral body wall and limbs ([Scaal and Christ 2004](#)).



Bonus e-book images

Fig. 44.15 Fetal myelomeningocele. Upper panel: three-dimensional ultrasound maximum transparency mode shows the spinal defect in the coronal (**A** and **B**) and oblique (**C**) views. Lower panel: three-dimensional ultrasound surface rendering mode shows the spinal defect in the sagittal (**A** and **B**) and coronal (**C**) views.

Fig. 44.16 Fetal magnetic resonance image shows a lumbar myelomeningocele in the sagittal section.

Fig. 44.17 The metameric structure of the myotomes of a 5-day-old chick embryo, visualized by whole-mount immunohistochemistry with an antibody staining for myosin heavy chain.

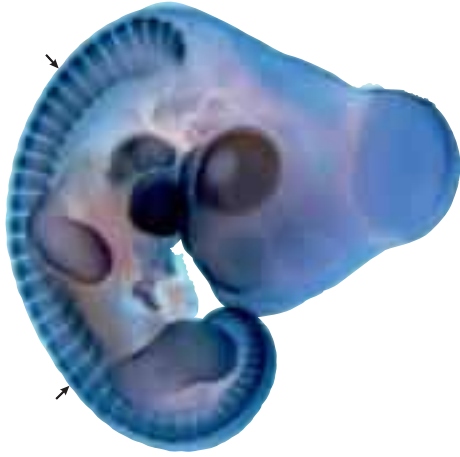


Fig. 44.17 The metameric structure of the myotomes (arrows) of a 5-day-old chick embryo, visualized by whole-mount immunohistochemistry with an antibody staining for myosin heavy chain.

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